Transmitter FFE design and simulation

HW8 - Introduction to Signal Integrity and Electromagnetic Compatibility

This report designs and evaluates a transmitter feedforward equalizer (FFE) for the supplied 50 cm stripline channel. All five required configurations are recomputed from the same random bit sequence. The post-FFE transmitter waveform is constructed explicitly and propagated through the channel by sectional convolution. The equivalent FFE frequency response is shown only for reference.

Model and parameters

Quantity	Setting
Channel	Supplied HW7_stripline_50cm.s2p, 50 ohm reference
Bit rate / modulation	40 Gb/s, binary NRZ
Unit interval	25 ps
Source bit levels	0 V and 2 V before FFE
Source-to-load transfer	H(f) = S21(f) / 2, applied once
Rise and fall duration	5 ps, 0 to 100%, for every transition size
Time sample spacing	0.5 ps, 50 samples/UI
FFE tap spacing	1 UI
Kaiser parameter	alpha = 0.5; beta = pi times alpha
Retained causal channel FIR	20 ns
Random sequence	65,536 measured bits; seed 0; guard bits excluded

Calculation sequence

Read the Touchstone frequency grid and complex S21. Apply the matching factor and Kaiser window, construct a Hermitian spectrum, and obtain a real sampled channel impulse response. Compute the single-bit response (SBR), sample its cursors one UI apart, solve the finite zero-forcing system, and normalize the taps.

Convolve the binary voltage levels with the taps. At each bit boundary, interpolate from the previous level to the new level over exactly 5 ps. Convolve this actual time waveform with the channel FIR in zero-padded sections, adding overlapping tails. Reuse the same bit pattern for every configuration.

Tap design and numerical results

Index channel cursors relative to the unequalized SBR maximum. For taps indexed from -Npre through Npost, use the following finite design system:

```
A[i,j] = a[r_i - n_j]         (unselected cursors set to zero)
A c_raw = e_main
c = c_raw / sum(abs(c_raw))
```

The main tap is at vector position Npre. A causal implementation stores the leftmost tap first and adds Npre UI of latency. A precursor does not require access to an unavailable future bit.

Case (pre / post)	Coefficients, left to right	Design residual (V)
1 (0 / 0)	+1.000000	0
2 (1 / 0)	-0.099584, +0.900416	5.55e-17
3 (0 / 1)	+0.687143, -0.312857	5.4e-18
4 (1 / 1)	-0.070629, +0.638611, -0.290760	2.78e-17
5 (1 / 2)	-0.071021, +0.642151, -0.276095, -0.010733	1.39e-17

The design residual measures A c minus the normalized target at the **retained design cursors**. It is not the residual of the full physical channel. Omitted channel cursors can contribute at these same times, and normalization reduces the main-cursor voltage from the unnormalized unit target.

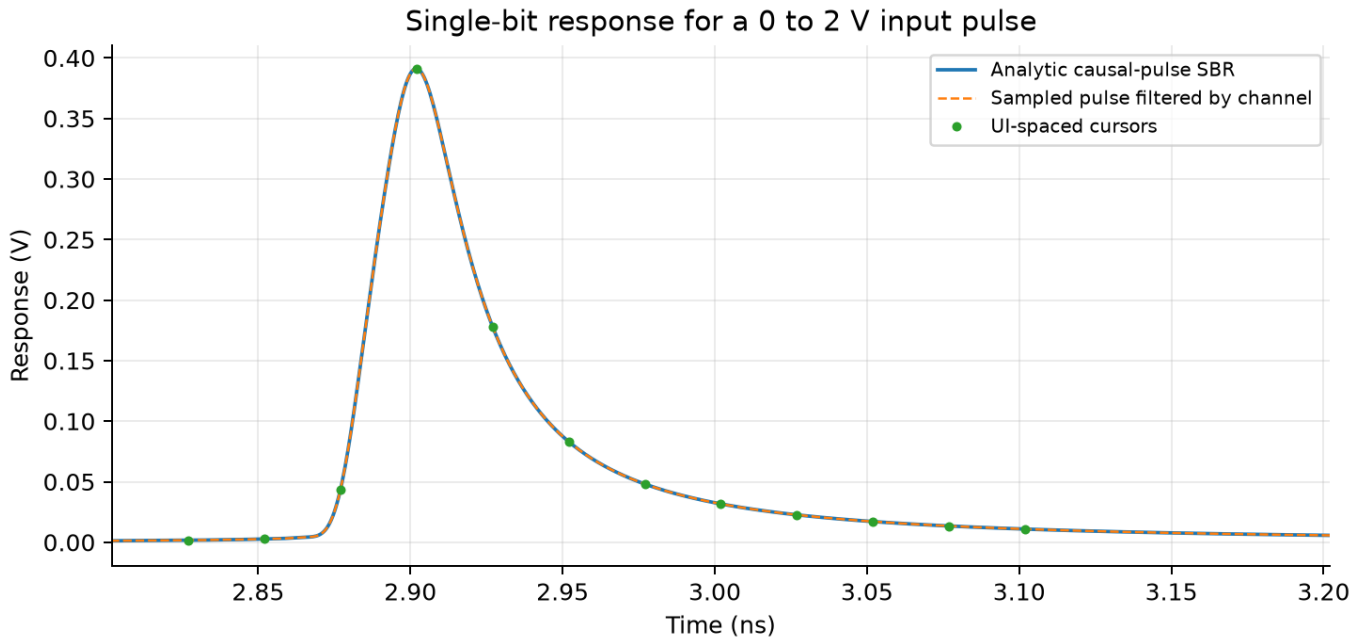
For example, case 4 retains a maximum selected off-main contribution of 5.871 mV at the original design phase. The absolute sum of all other UI-spaced analytic cursors at its chosen sampling phase is 69.380 mV. These physical-response quantities remain much larger than the finite matrix residual.

Case	SBR main (mV)	Empirical gap (mV)	Tx swing (V)
1	391.198	-93.407	2.000000
2	334.503	-38.655	2.000000
3	258.214	186.829	2.000000
4	226.692	180.814	2.000000
5	228.390	175.597	2.000000

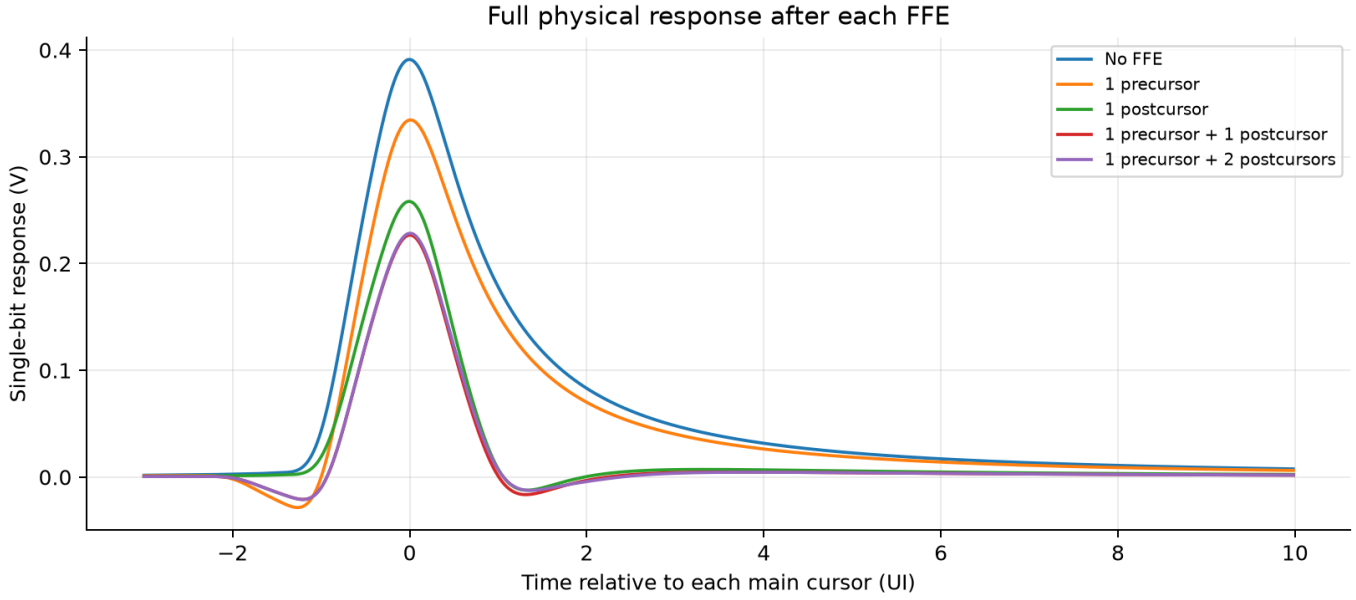
The empirical gap is the minimum sampled high voltage minus the maximum sampled low voltage in this finite, noiseless random run, at the phase selected by each equalized SBR. A negative value means the two observed distributions overlap. It is neither a BER estimate nor a 3-sigma eye-height measurement or a guarantee for every possible bit sequence. No gain or vertical offset is fitted between cases. Source swing is bounded by the absolute-sum tap constraint.

For this recorded run, case 3 has the largest empirical gap, 186.829 mV. Cases 1 and 2 have overlapping sampled distributions. The finite ZF system targets cancellation, not direct maximization of this gap; increasing the tap count does not make the gap improve monotonically.

Channel response in the time domain

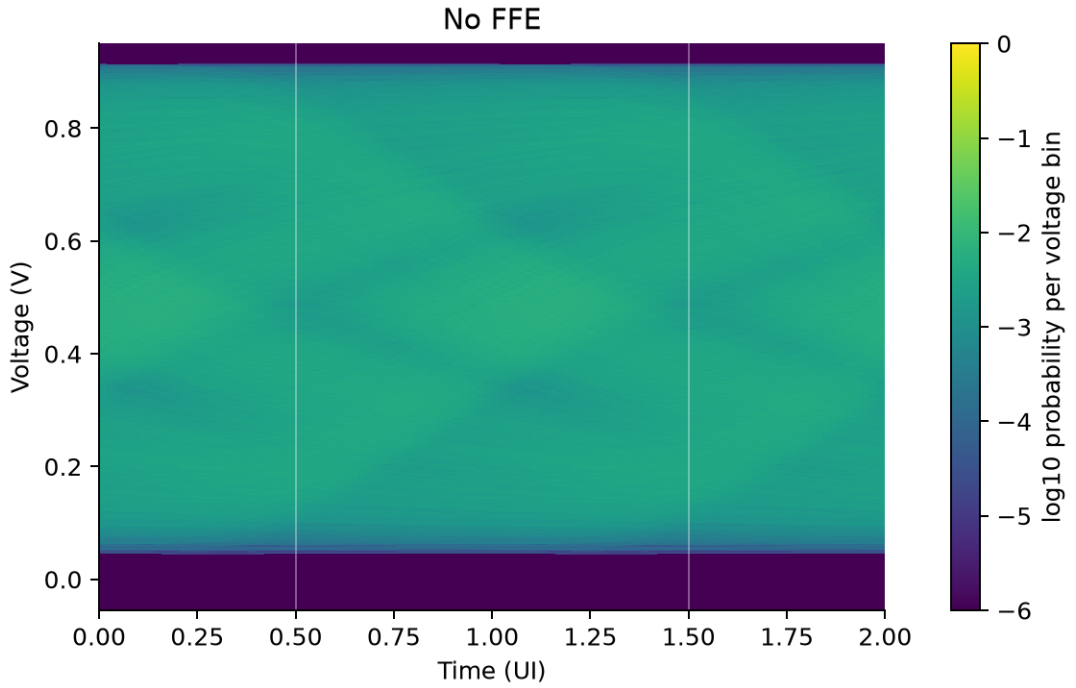


Unequalized single-bit response and UI-spaced cursor samples. The pulse phase uses Tb + Tr, giving a causal trapezoid.

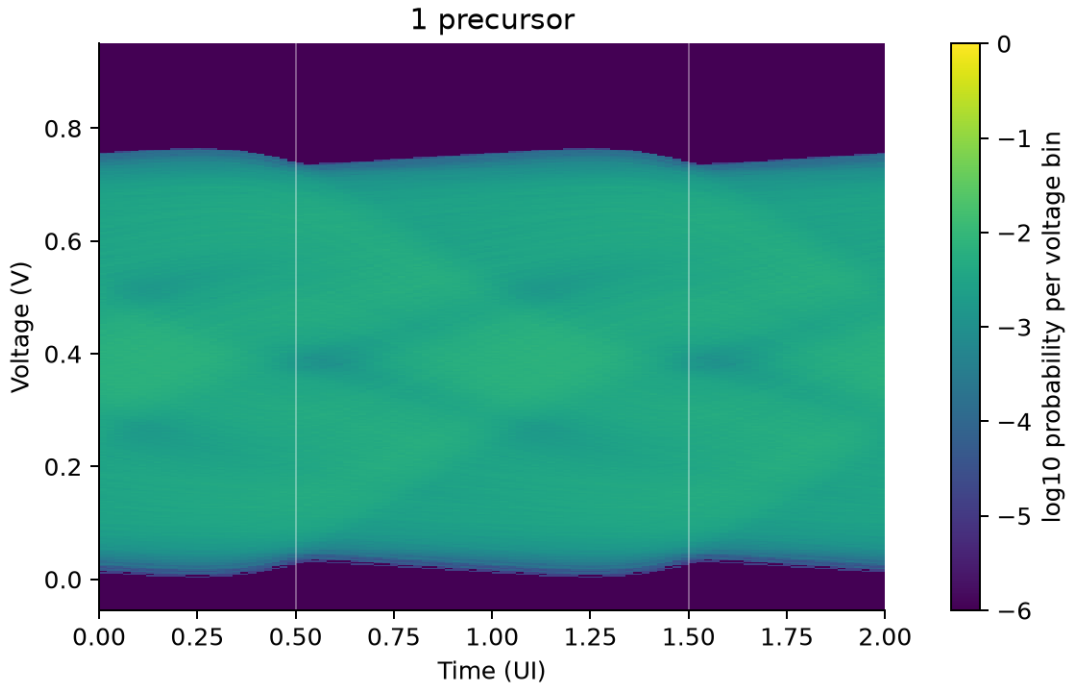


Full equalized SBRs. Finite tap designs reduce selected ISI, but residual precursor and postcursor contributions remain.

Eye diagrams: cases 1 and 2

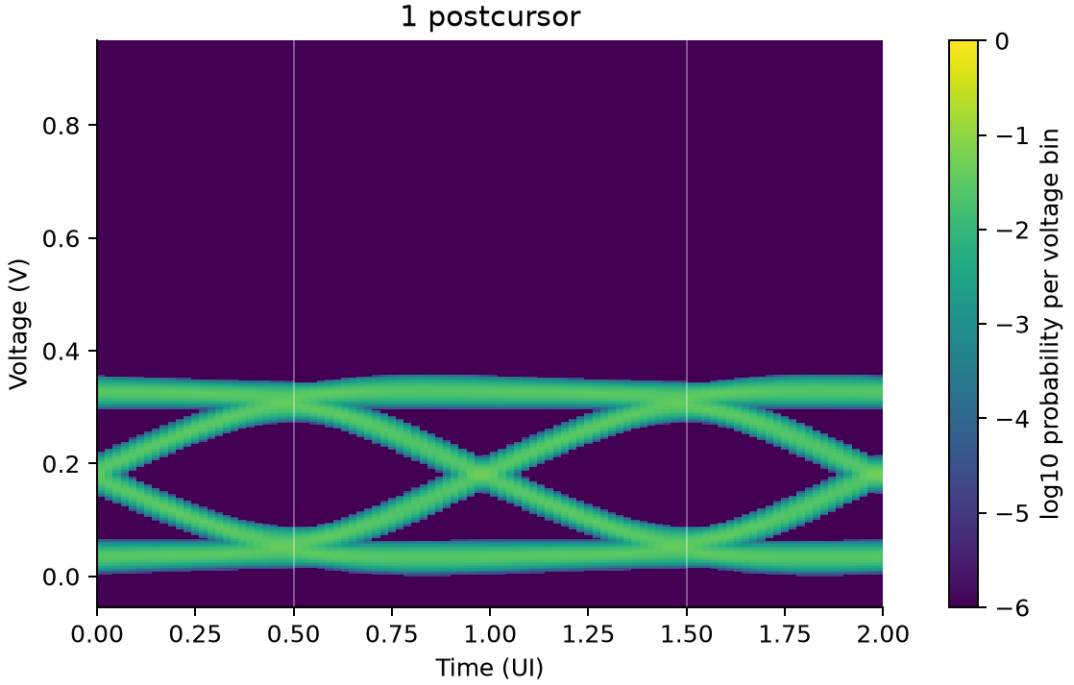


Case 1: unequalized reference. All eye plots use voltage increasing upward and two repeated UI of the phase-folded probability histogram.

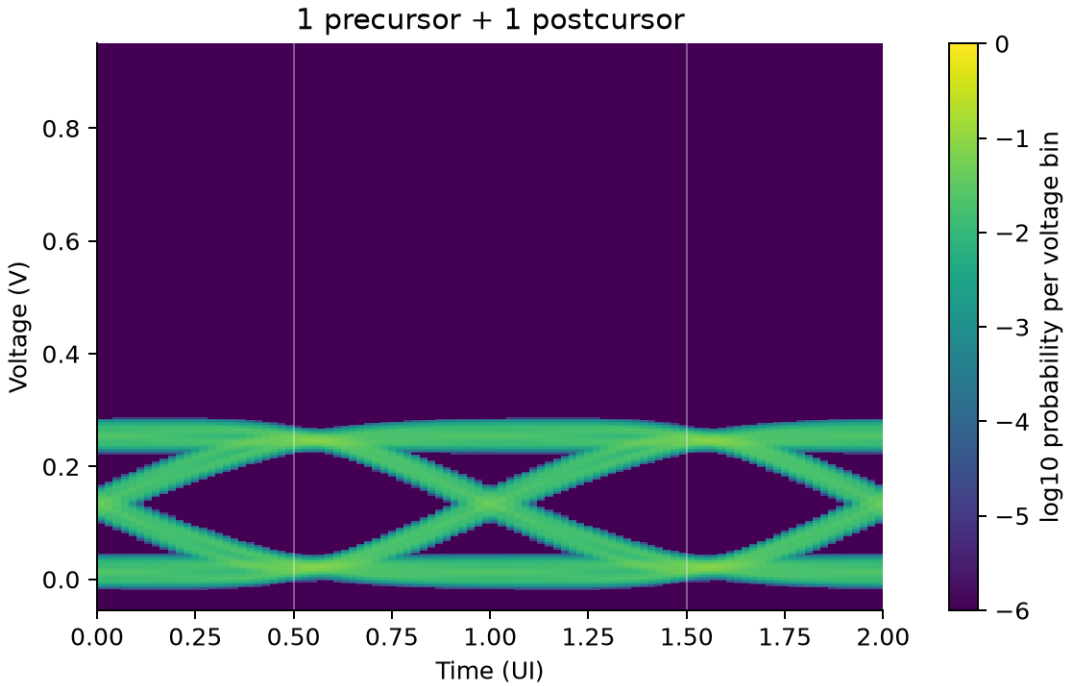


Case 2: one precursor tap and no postcursor taps. The main cursor is aligned consistently using this case's SBR.

Eye diagrams: cases 3 and 4

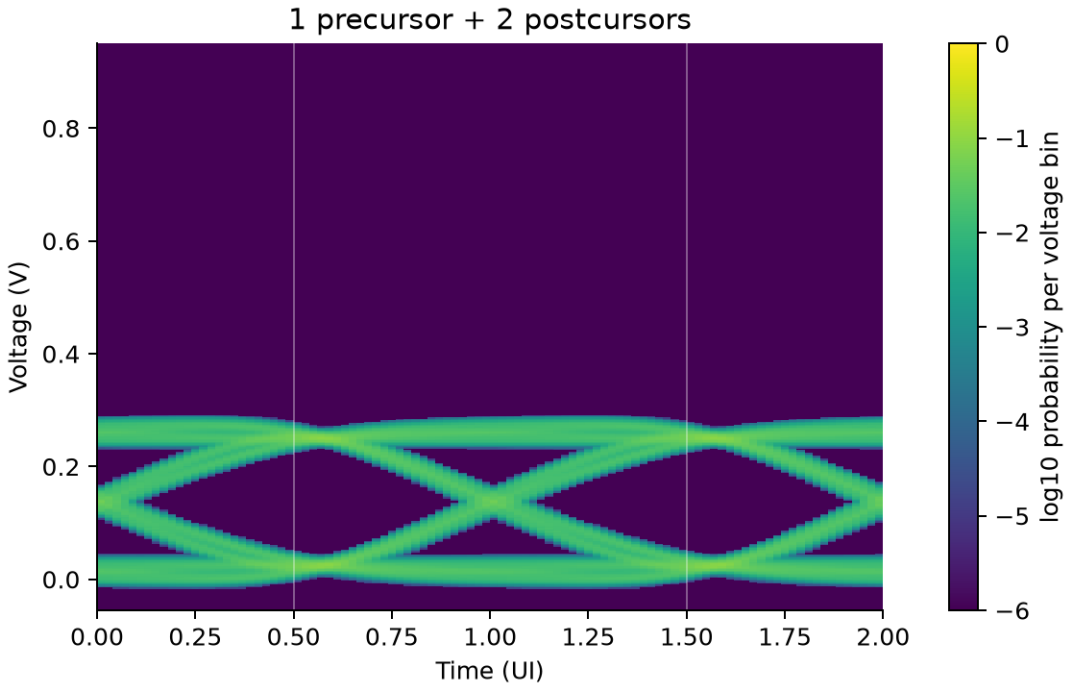


Case 3: one postcursor tap and no precursor taps. Transmitter de-emphasis reduces low-frequency source voltage under a fixed swing budget.

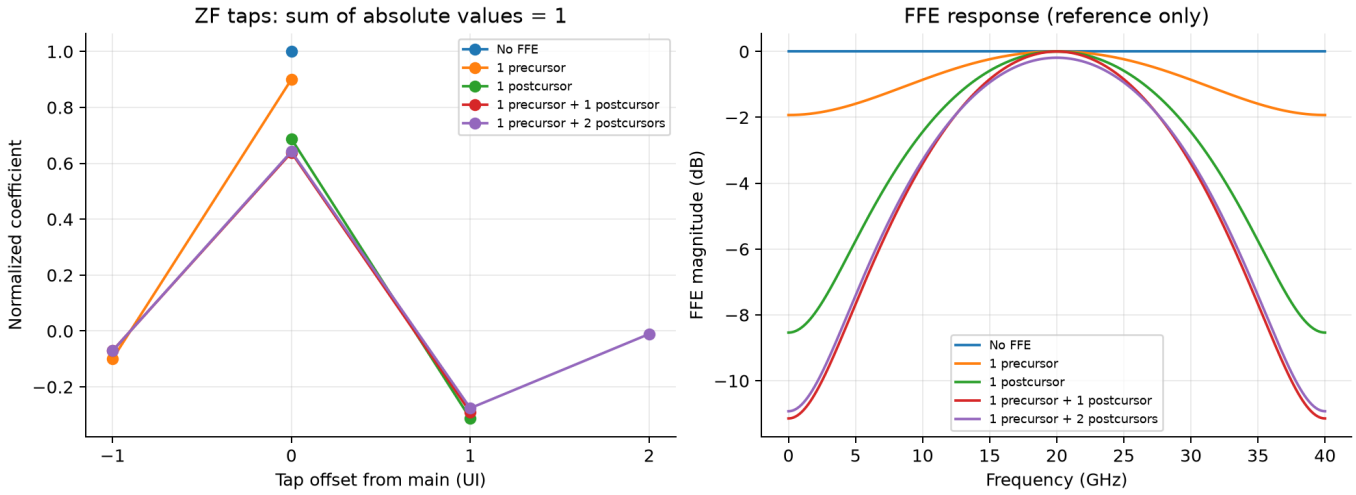


Case 4: one precursor and one postcursor. Its two transmitter snapshots and two receiver snapshots follow.

Eye diagram: case 5 and frequency reference

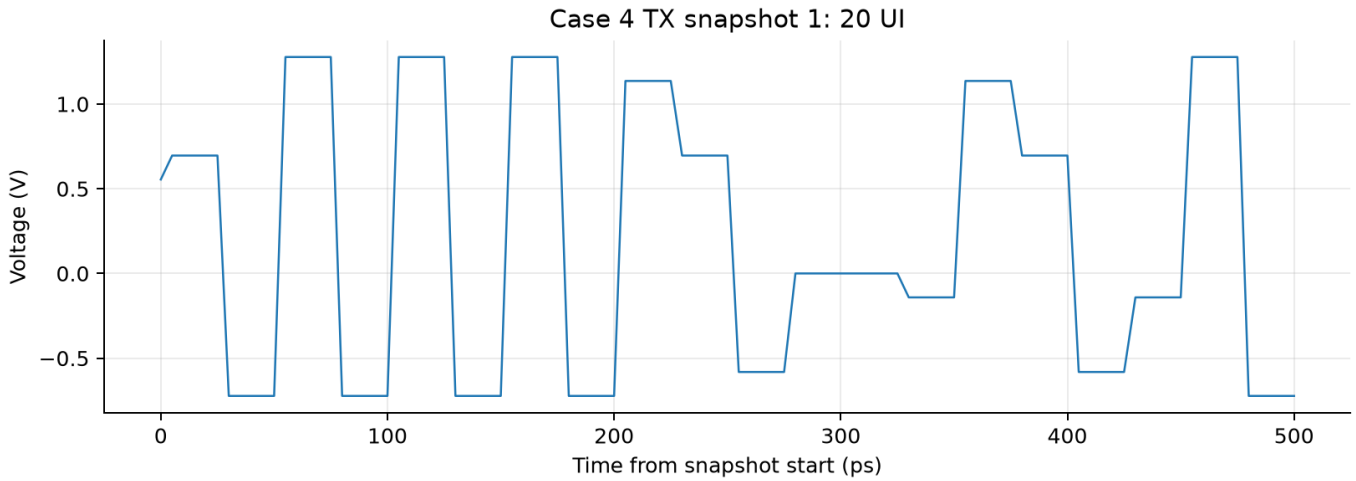


Case 5: one precursor and two postcursors. More taps address more selected cursors; they do not guarantee the largest sampled eye opening.

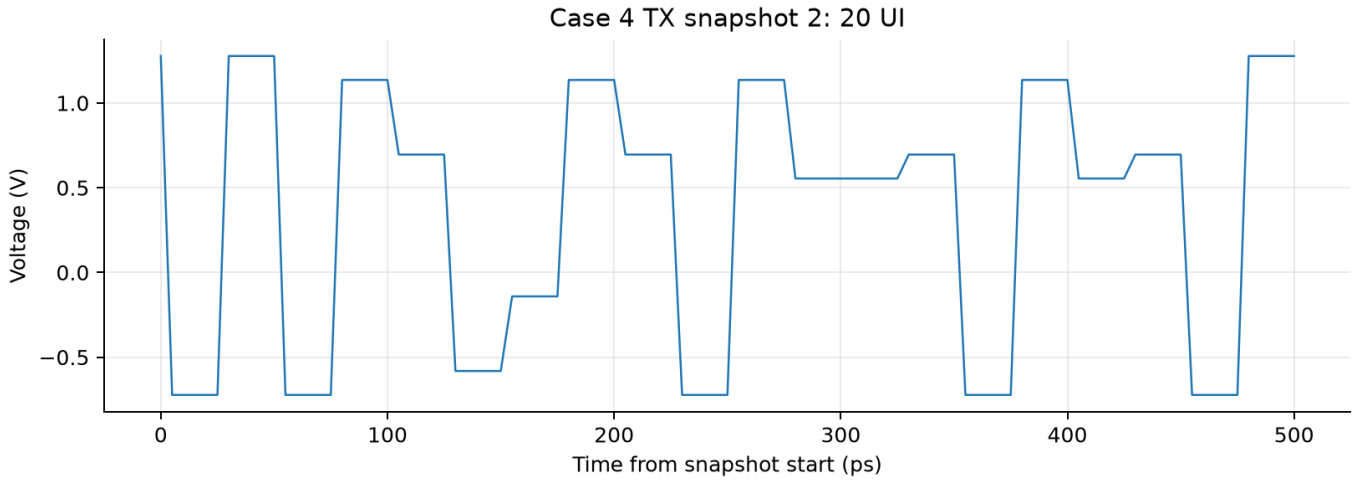


Normalized tap coefficients and equivalent FFE frequency responses. The waveform calculation uses explicit transmitter levels followed by channel convolution.

Case 4: transmitter waveforms

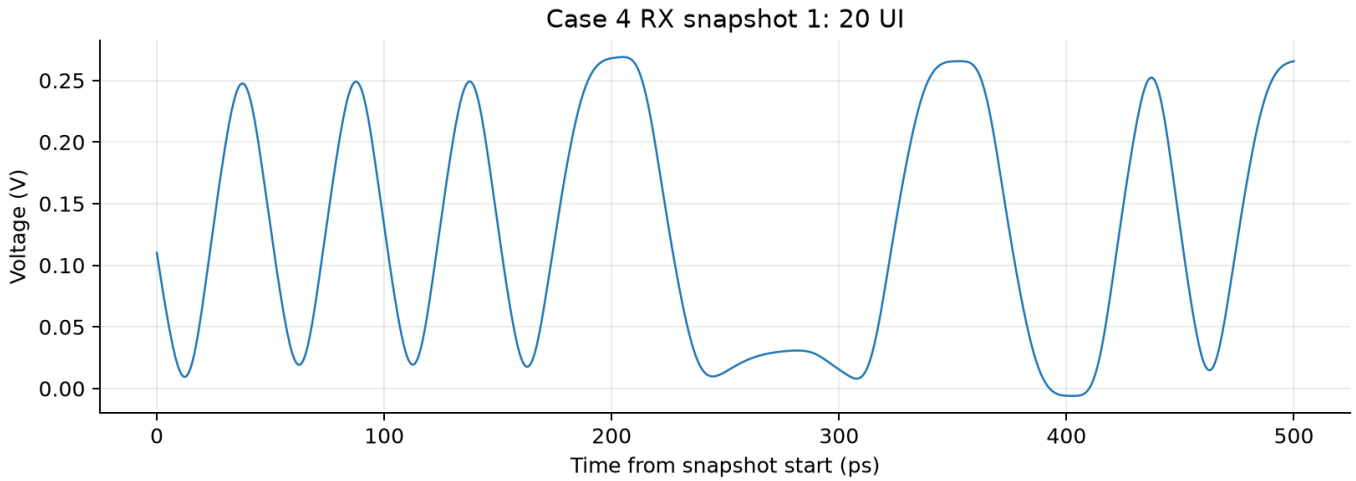


Transmitter snapshot 1. All level transitions use the same 5 ps duration, including transitions between intermediate FFE levels.

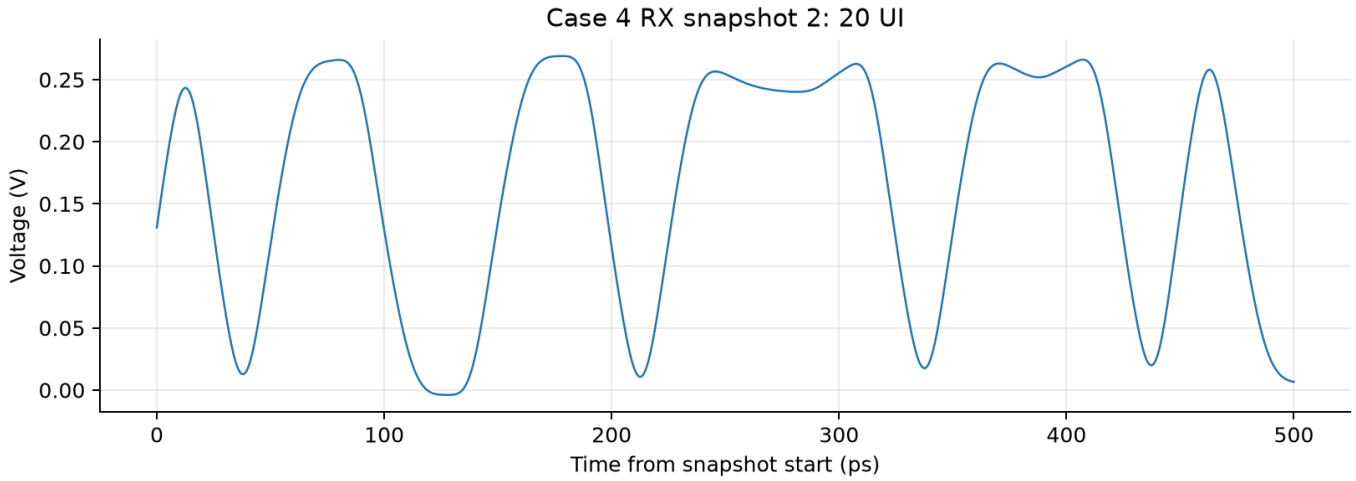


Transmitter snapshot 2, from a second part of the same deterministic random sequence. These two views satisfy the required Tx snapshots.

Case 4: receiver waveforms



Receiver snapshot 1. Time is relative to the recorded snapshot start, and all voltages retain the simulated source-to-load scale.



Receiver snapshot 2, from a second interval. These two views satisfy the required Rx snapshots.

Tx and Rx axes each start at their own local zero. Each paired Rx slice starts 2.913000 ns after its Tx slice, using the SBR-derived sampling latency minus half a UI. Their absolute start samples, start times and source payload indices are recorded in summary.json; the four CSV files preserve the plotted samples. The local zeros do not denote simultaneous physical times.

Verification and interpretation

Numerical check / recorded quantity	Value
Sectional vs direct convolution, max error (V)	2.32934e-21
Analytic vs sampled pulse, max error (V)	4.7629e-05
Analytic vs sampled pulse, RMS error (V)	5.89063e-07
Analytic vs sampled pulse, error / peak	0.000121752
Sampled DFT vs folded linear convolution, max error (V)	1.66533e-16
Periodic vs unfurled pulse, max error before 20 ns (V)	1.22307e-05
Main-window sampled pulse error at 0.5 ps (V)	4.7629e-05
Main-window sampled pulse error at 0.25 ps (V)	1.1903e-05
Refined time spacing (s)	2.5e-13

The independent regression run recorded **46 passing tests** (34 for HW8 and 12 existing regression tests), with 0 failures and 0 errors. It covers transfer-function scaling, waveform construction, tap indexing, section boundaries, eye normalization and existing repository analyses. The recorded evidence is in results/test_results.json.

The analytic SBR and the sampled finite-ramp convolution use different continuous-time versus discrete-time approximations. Their measured discrepancy is retained in the verification results. Zero filling above the measured bandwidth, Kaiser tapering, and finite causal impulse truncation also limit accuracy. A histogram with no added noise or jitter cannot establish link BER.

The finite 20 ns FIR has DC gain 0.483434557, compared with the source-to-load DC value 0.484733287; the relative DC deficit is 0.2679%. It is recorded rather than removed by an amplitude renormalization. The refined pulse check reduces the sampling discrepancy; it does not independently verify the random-data eye.

The optional ADS ChannelSim exercise is not run here. No numerical measured or ADS waveform reference is supplied for this HW8 data set, so the checks establish numerical consistency of the stated model, rather than an independent hardware or circuit-simulator match. Tx FFE emphasizes transitions by sacrificing steady-state amplitude under the swing constraint; an open eye need not have a larger absolute signal level.

Reproduce

```
python -m pip install -r requirements.txt
python homework/hw08/verify_inputs.py
python homework/hw08/analyze.py
python -m unittest discover -s tests -v
python -m pip install -r homework/hw08/report/requirements.txt
python homework/hw08/report/build_report.py
```

The supplied assignment and incomplete template are retained under instructions/. The channel and legacy Touchstone helper are supplied references; analyze.py is the rewritten executable solution. The original student report and source are not published. The report, PDF author field, figures and generated numerical data contain no student name, student number, email address or local machine path. Source-byte hashes are recorded in instructions/manifest.json.

Recorded runtime packages: python 3.14.4, numpy 2.5.0, scipy 1.18.1, matplotlib 3.11.2.